Emotion Recognition from Audio

Using Machine Learning Techniques

A Comparative Study

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Guide

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Introduction

Background of Problem

- Speech is the most natural way humans express emotions
- Emotions are conveyed through pitch, tone, energy, and speech rate
- Speech Emotion Recognition (SER) automatically detects emotions from audio signals
- Key features used for detection:
  - MFCC (Mel-Frequency Cepstral Coefficients)
  - Spectral Features
  - Zero Crossing Rate (ZCR)

Real-World Applications

- Customer service call analysis
- Mental health monitoring
- Human–computer interaction
- Virtual assistants & smart devices
- E-learning platforms
- Healthcare & security systems

Problem Statement

Why is Emotion Recognition Challenging?

- Emotions vary across different speakers and genders.
- Background noise reduces audio signal quality
- Some emotions (e.g. Fear & Sadness) have very similar acoustic patterns
- Limited availability of labeled emotional speech datasets

Input / Output

Input: Audio speech signal (WAV format)
Output: Happy, Sad, Angry, Fear, Disgust, Neutral

ML Task Type

Multi-Class Classification

Motivation for Topic Selection

Why This Topic

- Emotion detection improves human–computer interaction
- Makes AI systems more intelligent and empathetic

Real-World Need

- Helps analyze customer feedback automatically
- Monitors mental health through voice signals

Research Gap

- Existing systems struggle in noisy environments
- Similar emotional patterns cause misclassification

Social & Industrial Impact

- Useful in call centers, healthcare, and AI assistants
- Supports emotion-aware security systems

Academic Interest

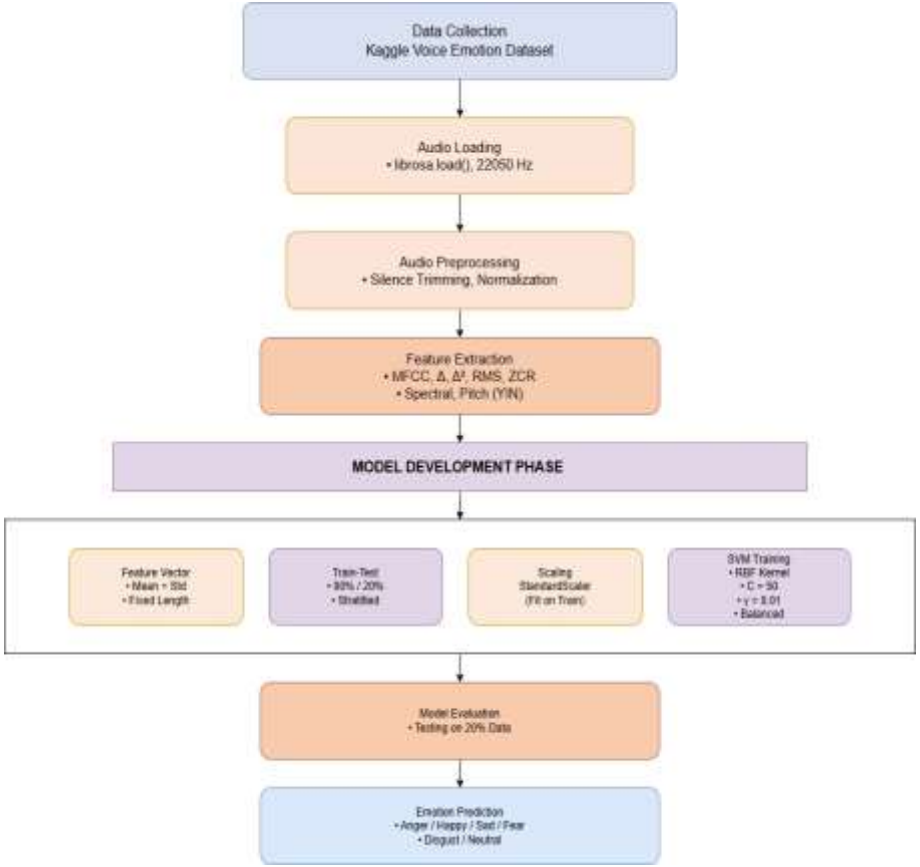
- Combines ML, signal processing, and AI
- Practical application of classification algorithms

This project bridges the gap between AI technology and real human emotional communication.

Literature Review — 15+ Papers Reviewed

Study	Method	Limitation
MFCC, Energy, Pitch	Handcrafted acoustic features + classical ML classifiers	Limited capture of complex emotional variations
MFCC, Pitch, Energy, ZCR	KNN, Naïve Bayes	Sensitive to noise and speaker variability
Spectral features, ZCR	Random Forest	Reduced performance in high-dimensional feature space
MFCC, Spectral, Pitch, Energy, ZCR	SVM (RBF Kernel)	Limited cross-dataset generalization in real-world testing

Methodology: System Pipeline



Dataset Description

Dataset Details

- Source: Kaggle — Voice Emotion Classification Dataset
- Created by: S. Deogade (publicly available)
- Format: WAV audio files
- Sampling Rate: 22,050 Hz
- Balanced dataset across all 6 emotion classes
- Various speakers — diverse voice types & genders

6 Emotion Classes

- Anger
- Happiness
- Sadness
- Fear
- Disgust
- Neutral

Challenges

01 Background Noise

Real-world recordings contain noise that hides emotional features in the voice signal

02 Similar Emotion Patterns

Fear & Sadness, Anger & Happiness share very similar acoustic features

03 Speaker Variability

Same emotion sounds different across various speakers, genders, and accents

04 Limited Labeled Data

Annotating emotional speech is time-consuming and expensive

Conclusion

This research explores machine learning approaches for detecting emotions from speech signals, enabling intelligent systems to understand human emotional states.

Human-Computer Interaction

Speech emotion recognition improves how machines understand and respond to human emotional states in real time.

Wide Applications

Applications span customer service (call analysis), healthcare (mental health monitoring), and education (adaptive learning).

Future Work

Future work includes developing real-time emotion detection systems and multimodal approaches combining speech with facial expressions.

References

[1] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[2] V. N. Vapnik, Statistical Learning Theory. Wiley-Interscience, 1998.

[3] D. Ververidis and C. Kotropoulos, "Emotional speech recognition: Resources, features, and methods," Speech Communication, vol. 48, no. 9, pp. 1162–1181, 2006.

[4] B. Schuller, S. Steidl, and A. Batliner, "The INTERSPEECH 2009 Emotion Challenge," Proc. INTERSPEECH, pp. 312–315, 2009.

[5] A. Mehrabian, "Communication without words," Psychology Today, vol. 2, no. 4, pp. 53–56, 1968.

[6] K. S. Rao et al., "Emotion recognition from speech," IJCSIT, vol. 3, no. 2, pp. 3603–3607, 2012.

[7] P. Sharma et al., "SER using kernel sparse representation-based classifier," EUSIPCO, pp. 374–377, 2016.

[8] S. R. Bandela and T. K. Kumar, "Stressed speech emotion recognition using Teager Energy + MFCC," ICCCNT, 2017.

[9] A. A. Zamil et al., "Emotion detection using voting mechanism on classified frames," ICREST, IEEE, 2019.

[10] L. Kerkeni et al., "Automatic speech emotion recognition using machine learning," 2019.

[11] S. Deogade, "Voice Emotion Classification Dataset," Kaggle. Available: kaggle.com/datasets/sdeogade/voice-emotion-classification

[12] B. Schuller et al., "Recognising realistic emotions in speech," Speech Communication, vol. 53, pp. 1062–1087, 2011.

[13] F. Eyben et al., "OpenSMILE: The Munich open-source audio feature extractor," ACM Multimedia, 2010.

[14] C. M. Lee and S. S. Narayanan, "Toward detecting emotions in spoken dialogs," IEEE Trans. Speech & Audio, 2005.

[15] C. Busso et al., "Fundamental frequency analysis for emotion detection," IEEE Trans. Audio, Speech & Lang., 2009.